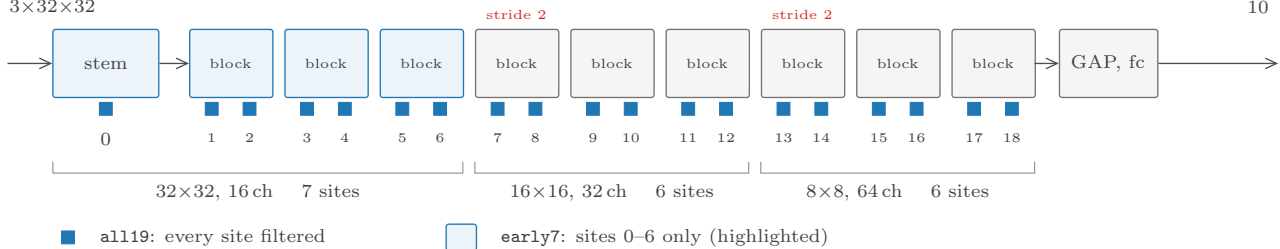


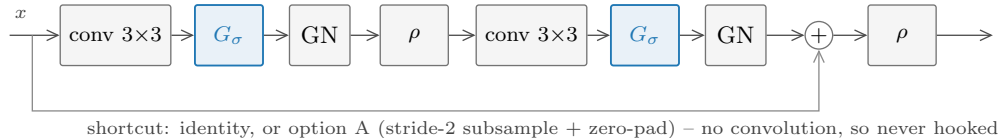
### A 19 insertion sites: every main-path $3 \times 3$ convolution

One forward hook per  $3 \times 3$  convolution, in module order. Site 0 is the stem; sites 1–6 are stage 1, which is exactly the **early7** set. 10



### B Inside one basic block

The hook rewrites the *output of the convolution*, before GroupNorm and before the ReLU.



### C How $\sigma$ is set at each site

The stored level is rescaled by the site's relative width, so a level means the same fraction of a feature map at every depth.

$$\sigma_l(e) = q_l(e) L[e][l], \quad q_l(e) = \frac{\text{width at site } l \text{ now}}{\text{width at site } l \text{ at } 32 \times 32}$$

R32:  $q_l = 1$  everywhere.

input.\* reduction to  $r$ :  $q_l = r/32$  at all 19 sites.

stem.\* reduction to  $r$ : site 0 keeps  $q_0 = 1$ , the other 18 get  $r/32$ .

From epoch 21 of 30,  $\sigma = 0$ : an exact identity, and the plain network trains to the end.